

TARUN SINGH GOHIL

Senior Python / Full-Stack AI Engineer

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PROFESSIONAL SUMMARY

Python-focused Full-Stack Engineer with 5+ years of professional experience building production web applications, enterprise SaaS products and AI-enabled workflows. Hands-on experience across Python, Django, Django REST Framework, FastAPI/Flask, REST APIs, authentication and authorization, ORM/data modelling, SQL/NoSQL databases, background processing and third-party integrations. Strong full-stack ownership across React/TypeScript frontends, API contracts and end-to-end feature delivery. Practical Generative AI exposure includes OpenAI GPT integrations, prompt engineering, RAG-style retrieval, embeddings/vector search concepts, OCR/document understanding and speech-to-text workflows. Experienced with AWS, Docker, CI/CD fundamentals and Agile/Scrum delivery, with a focus on readable code, reliable APIs, debugging and production problem solving.

PROFESSIONAL EXPERIENCE

Lead Developer — Python & Full-Stack | Growit.ai Technology Private Limited, Jaipur | Aug 2025 – Aug 2026

- Led development of a scalable, multi-tenant, AI-powered Hospital Information Management System (HIMS), working hands-on across Python/Django backend services, REST API integration and React frontend architecture.
- Built and integrated Python + Django REST Framework APIs for enterprise healthcare workflows, including serializers, viewsets, permissions, middleware, ORM queries, migrations, validation, pagination and filtering.
- Designed REST API contracts consumed by web clients, covering request/response payloads, JWT authentication, tenant-aware authorization, error handling and asynchronous client-server communication.
- Implemented permission-driven, role-based access and tenant-aware data isolation for a large set of business endpoints, helping keep hospital data scoped to the correct organization and user context.
- Contributed to AI/OCR-enabled workflows, connecting document data extraction and LLM-assisted capabilities with clinical, reporting and billing-oriented application flows.
- Delivered complex modules spanning Patient Management, OPD, IPD, Emergency, Laboratory, Radiology, Pharmacy, Billing, Inventory, Analytics, Reporting and NABH-related compliance workflows.
- Modelled and optimized data operations across MySQL and MongoDB, including schema decisions, indexing, query tuning and caching for data-heavy screens and API flows.
- Built the enterprise React 19 / TypeScript / Redux Toolkit / Tailwind frontend with reusable components, dashboards, data tables and role-based interfaces integrated with Python and Node.js services.
- Owned Git branching, pull requests, code reviews, release coordination, Docker-based development environments and deployment support while collaborating with product, QA, design and backend engineering teams.
- Worked across Python/Django and Node.js/Express services, aligning frontend and backend contracts and troubleshooting API, authentication, data and deployment issues through the development lifecycle.

Software Engineer | Formidium India Private Limited, Jaipur | Feb 2022 – Jul 2025

- Developed and maintained production-grade enterprise fintech/SaaS applications, working hands-on with Python for application logic, API development, data-processing workflows and backend-supported business features alongside React.
- Built Python/Django-based services and REST APIs for business workflows, including request validation, authentication, ORM-driven database operations and third-party system integrations.
- Wrote Python scripts and utilities for data extraction, transformation, validation and reporting, replacing repetitive manual steps with repeatable and testable automation.
- Collaborated on API contracts, response handling, pagination and performance tuning with backend teams, and debugged production issues across frontend, API and data layers.
- Developed reusable React/TypeScript components, custom hooks and Redux Toolkit / Context API state patterns for dashboards, operational tools and customer-facing products.
- Built analytics dashboards using Chart.js and Recharts to present KPIs, financial information and operational business metrics sourced from backend APIs.
- Managed and optimized 12+ production applications through caching, code splitting, lazy loading, API optimization and Webpack/build improvements to support responsive user experiences.

- Participated in Git-based development, code reviews, testing, Agile/Scrum ceremonies and CI/CD-supported release workflows across multiple product teams.

Web Developer | i3Techs, Jaipur | Nov 2020 – Feb 2022

- Contributed to full-stack-oriented development workflows by working across frontend interfaces, API integrations, client-side functionality and production deployment requirements, building a strong foundation for subsequent Python and backend engineering responsibilities.
- Developed and maintained 50+ production websites and web applications using HTML5, CSS3, JavaScript, Bootstrap and React.js, contributing to end-to-end application development with API-driven frontend implementations and backend-integrated workflows.
- Built reusable and responsive UI components from PSD/Figma designs, focusing on clean architecture, semantic HTML, accessibility-aware implementation, cross-browser compatibility, maintainability and performance.
- Worked on API-integrated web applications, handling frontend-to-backend data flows, request/response integration and dynamic business data while collaborating on application requirements and delivery.
- Developed and maintained WordPress/Elementor-based business websites and client-facing applications with attention to usability, responsive behaviour, SEO-aware structure, maintainability and production quality.

Web Development Trainee | i3Techs, Jaipur | May 2019 – Jul 2019

- Completed hands-on training in web application development, responsive UI/UX implementation, frontend architecture fundamentals and production-oriented development practices.
- Worked with HTML, CSS, JavaScript, responsive design principles and Adobe XD/Photoshop while learning how to translate design requirements into structured, maintainable web interfaces.
- Built foundational understanding of application development workflows, debugging, browser-based development and collaborative delivery practices, establishing the base for subsequent full-stack engineering experience.

GENERATIVE AI & AI ENGINEERING EXPERIENCE

- Integrated LLM-powered features into production applications using OpenAI GPT models for document understanding, content generation, summarization and assisted data entry within business workflows.
- Designed and iterated system/role prompts, few-shot examples and structured JSON output contracts so model responses could be validated and consumed reliably by downstream application logic.
- Built RAG-style flows covering document ingestion, chunking, embeddings, similarity search and grounded response generation using retrieved domain context.
- Delivered OCR and computer-vision-assisted document pipelines to extract structured information from scanned reports and forms, reducing manual data-entry effort.
- Implemented speech-to-text workflows with Whisper / Faster-Whisper for transcription and voice-driven data capture scenarios.
- Exposed AI capabilities through Python REST services with request validation, error handling, retries, timeouts, usage controls and safe fallback behavior when model calls failed.
- Applied working knowledge of agentic patterns across LangChain, LangGraph, CrewAI and AutoGen, including tool calling, routing, state/memory and human-in-the-loop review steps.

SELECTED PROJECTS

Zonov — Multi-Tenant AI-Enabled Hospital Information Management System — Enterprise Healthcare / HIMS

Stack: Python, Django REST Framework, REST APIs, JWT, RBAC, MySQL, MongoDB, Redis, Docker, AWS, OpenAI/OCR, React 19, TypeScript, Redux Toolkit

- Built Python/DRF backend services and API integrations for a multi-tenant HIMS covering patient, OPD/IPD, emergency, laboratory, radiology, pharmacy, billing, inventory, reporting and compliance workflows.
- Implemented tenant-aware access control and permission-driven workflows while integrating AI/OCR-assisted document processing into application-facing business flows.

AI Document & Voice Assistant Workflows — GenAI / Document Intelligence

Stack: Python, FastAPI, OpenAI GPT, prompt engineering, embeddings & vector search, Whisper / Faster-Whisper, OCR, REST APIs

- Developed Python services that ingest documents, extract structured data through OCR, retrieve relevant context with embeddings and vector search, and generate grounded, schema-validated LLM responses.

- Added speech-to-text capture using Whisper / Faster-Whisper for hands-free data entry and transcription-oriented workflows, with API validation and application integration.

RetailX ERP — POS Billing, Inventory & Reporting — Enterprise Business Application

Stack: Python/Node.js services, REST APIs, MongoDB, React.js, TypeScript, Redux Toolkit, Tailwind CSS

- Delivered data-driven billing, inventory and reporting workflows with API integrations, role-based access patterns and reusable frontend architecture.
- Worked across business-facing screens and backend-connected flows where data consistency, responsive UI behavior and API handling were core requirements.

Charge My EV — EV Charging Platform — Web + Mobile Product

Stack: React Native, React.js, Node.js, Express.js, MongoDB, REST APIs

- Built customer-facing EV charging workflows and API-driven features across web and mobile surfaces, working with reusable UI patterns and backend integrations.

TECHNICAL SKILLS

Python & Backend: Python 3.x, OOP, abstract base classes, abstraction, encapsulation, inheritance, polymorphism, decorators, decorator chaining, generators, iterators, list/dictionary comprehensions, slicing, mutability, context managers, custom exceptions, typing/type hints, async/await, threading vs multiprocessing, scope and memory fundamentals, Pythonic clean-code practices.

Django / Django REST Framework: Django, Django REST Framework (DRF), application architecture, REST API development, serializers, viewsets, middleware, authentication, permissions, ORM queries, migrations, signals, validation, pagination, filtering, request/response handling and integration with React clients.

FastAPI / Flask: FastAPI async endpoints, Pydantic validation, dependency injection, auto-generated API documentation, lightweight service development, Flask-based utilities and internal services, Celery-style background and asynchronous task processing.

Generative AI / LLM / ML / NLP: OpenAI GPT, Azure OpenAI concepts, AWS Bedrock concepts, Google Vertex AI concepts, prompt engineering, prompt templates, system/role prompts, few-shot examples, structured JSON outputs, RAG workflows, document ingestion, chunking, embeddings, similarity/vector search, function/tool calling, token and cost optimization, LLM evaluation, guardrails, scikit-learn, NumPy, Pandas, NLP concepts, computer vision, OCR, Hugging Face, Whisper and Faster-Whisper.

Agentic AI: LangChain, LangGraph, CrewAI, AutoGen and Google ADK working knowledge; multi-agent workflow design, agent routing, tool/agent orchestration, memory and state handling, human-in-the-loop review and approval patterns.

Frontend: React 19, React.js, JavaScript ES6+, TypeScript, React Hooks, React Router, Next.js, Redux Toolkit, Redux, Context API, Zustand, TanStack Query, form and API state management, Tailwind CSS, Material UI, Bootstrap, HTML5, CSS3, responsive design, accessibility-aware implementation and Angular fundamentals.

Databases & Data: MySQL, PostgreSQL, MongoDB/Mongoose, Redis; SQL joins, subqueries, normalization, indexing, query optimization, transactions, schema and data modelling, ORM-based database operations and caching strategies.

APIs & Architecture: REST/JSON API design, API versioning, microservices concepts, JWT authentication, RBAC, multi-tenancy, pagination, validation, filtering, error handling, retries, timeouts, rate limiting, webhooks, WebSockets, GraphQL, third-party API integrations and payment integrations.

Cloud / DevOps: AWS Lambda, EC2, S3, CloudWatch, IAM basics, serverless concepts and concurrency limits, Azure and GCP fundamentals, Docker, Docker Compose, Linux, Git, GitHub, GitLab, CI/CD workflows, environment and release management and Vercel.

Testing & Quality: pytest, unittest, fixtures and mocking, API testing with Postman, Jest, React Testing Library, Cypress, Playwright, code reviews, debugging, performance profiling and production issue triage.

ENGINEERING PRACTICES, EDUCATION & CERTIFICATIONS

Architecture & Delivery: Modular backend services, REST-first design, reusable business logic, clear authentication/authorization boundaries, API contracts, Git branching and pull requests, release workflows, Docker-based environments, CI/CD fundamentals, cloud deployment support, Agile/Scrum, requirements analysis and end-to-end feature ownership.

Education:

B. Tech, Computer Science & Engineering — Global Institute of Technology, Jaipur | 2016–2020.
Senior Secondary (Science) — Anand Public School, Sawai Madhopur | 2016.